

City University of Hong Kong 2025-2026
Semester A CS3343 Software Engineering
Practice

Self Assessment Report
Group 19
Classic UNO Card Game

Conducted By:

Woo King Hei 58533231

Wong Ka Chung 58533673

Chan Pak Yuen 58535918

Chan Wing Ki 58111083

Ho Sung Hin 58010837

Wong Hei Leong 58537010



Woo King Hei 58533231

As the project manager of this group, I was responsible for assigning roles, hosting meetings, as well as making sure our progress is on track. Additionally, I also carried out bug testing from the users' perspective as well as the report documentation. I certainly have learnt a lot from this project, mainly from the following two aspects.

The first aspect is the project managing aspect. I learnt to assign roles according to my groupmates' different strengths, such as assigning groupmates with stronger coding skills for developing the more crucial components of the UNO game. Apart from this, throughout the project, I improved my communication skills when hosting meetings. I am now more capable of expressing views and suggesting directions for the team effectively without causing any misunderstanding due to wrong wording, omitted details, etc. Furthermore, I also learnt better time management. During meetings, I set internal deadlines for each phrase and give different time limits depending on the work required for the phrase. This greatly prevented deadline fighting and our group was able to complete the UNO project in time without compromising quality. When setting deadlines, I also made sure to provide buffer periods to prevent the whole process being too stressful and rushed.

The second aspect is the practical aspect in the project. I carried out vigorous bug testing on the UNO game, and reported any bugs or feature requests via BigZilla. The biggest challenge of this task is trying to figure out any edge cases that should be tested out, such as when the deck is empty, deliberately clicking the deck to draw many cards, and attempting actions on other players' turn for example calling UNO and playing cards. Although these cases are hard to find and test out, these cases often turned out to be the scenarios where the code may dysfunction and produce bugs. After this project, I learnt to better identify edge cases by deliberately trying to "break the rules" and not perform actions that the program intends the user to perform.

Overall speaking, I gained a lot from this project as a project manager as well as an actual developer. The results of this project would not have been possible without my teammates and I cannot thank them enough.

Wong Ka Chung 58533673

During this project, I am responsible for initiating the stage of the code, preparing part of the test cases for the finished code, making the class diagram, and the sequence diagram. At the start of the project, the codebase is completely empty. I try to write the code for the UI of the view component and create the UNO controller as the control system to initialise the program. The game logic is left to be finished by my groupmates. Meanwhile, I am in charge of most test cases for those components. I need to ensure full coverage for the whole codebase. Hence, I write each test case for each unit file, then integrate further to the abstract class, and at last to the whole system. This allows my teammate to find any bug. When we finalise the code, I am responsible for making the class diagram and some sequence diagrams to showcase the whole structure of the code and how the function flows. Then, I will present them and make changes under some change requests from my groupmates.

To conclude, I act as a supporter of the project. When my groupmates need help, I will support them and finish the tasks as requested. I learned that any resource applied to any work will affect the whole project at the right time. Even though I am a support but not the main programmer for the project, I am still able to contribute at the right moment. I also act as a test engineer in the project, who is responsible for writing the code for inspection, in order to test any problems in the system. This has made me learnt the importance of test cases to inspect the code to make the code error-free or within a tolerance level, reducing the time cost of solving errors.

Wong Hei Leong

As a Coordinating Project Manager in this project, I gained a deeper understanding on the importance of project management. Comparing with CS3342 which merely draw diagrams and simply make some UI photos, we really needed to implement the program of our product which totally increased the difficulty of time management since you cannot force to put all the concepts within the diagram, but genuinely implemented in our development and generate the diagrams according to our program. Therefore, it takes time to coordinate the progress and the communication of our team members. My duties were to arrange meetings, monitoring our products' quality and reports. Due to the difference of our schedules from the internships, the face-to-face meeting time we can arrange is only Friday, which is the day at CityU. However, as time goes by, I found that there is a lack of time to communicate since it is easy to have conflicts due to the expectations of the deliverables and therefore we need additional time to coordinate the deliverables of reports and the programs. For instance, it took time to reach consensus on how to apply SOLID design principles to our program and the testing approach of our program. If we cannot reach the same idea, it may negatively affect the product's quality. Therefore, we are starting to rely on messaging app to have a real time call which is also a method of communication such as Discord and we can easily have a look at everyone's progress by screen sharing.

Moreover, I always encourage team members to deliver their difficulties and bad messages as one of the purposes of meetings since it always takes time to record and resolve the issue such as the bug report documentation. If I always effectively control our communications as an optimal flow, I can not only make good progress, but also build a good relationship with team members, which can motivate them to work on the project more. In conclusion, I think I have gained a lot in project time management and communication management. It is hoped that I can make good use of the experience of this project to unleash my potential in my internships and future workplace.

Chan Pak Yuen

During this project, my duties were setting up test cases for some classes, making charts and sequence diagrams and the demonstration video. I would like to thank my supportive group mates, they gave me advice when I am not familiar with the codes or tasks. Effective communication is really important for the project as me and my teammates are having an internship at this moment. Effective communication can reduce the chance of misunderstanding each instruction and doing unnecessary tasks.

I also learnt how to make reasonable test cases. Not to rush or put too many test items in a single Junit test. As I mentioned we are having an internship so time management is a major difficulty for me. It is hard to maintain work life balance. In order to finish the tasks on time. I used some time during the lunch break to do the tasks.

Chan Wing Ki 58111083

As a first-time analyst programmer, I understand that this role is not only about coding. In this project, my main duties also included analysing the UNO game rules to define requirements and scope, designing the whole UNO game system structure and ensuring the code and test cases are working and clean.

At the beginning of the project, I discussed and wrote down the Uno game flow, rules and some game edge cases with my teammate. However, I found later in the project that I had not considered some cases, which led to some bugs and issues in the programming. For example, when all deck cards are drawn, the game should end immediately. This makes me spend more time solving these bugs and issues. Also, I realized that I was not careful enough when I designed the program.

Moreover, when I explained my idea or concept to my other teammates, I found that they sometimes didn't clearly understand what I was talking about. So, I realized that to make the communication clearer, I can directly use some graphs or show an example when I am explaining. After I did it, it actually improved my communication.

To ensure the code is clear and maintainable, I also checked the code after we finished each method or test case. I applied the different code refactoring skills I learned in the lecture, like extracting the method to handle duplicated methods and reducing the method length. Although I spent extra time on code refactoring, it helped me and others more easily read the code and modify it.

Finally, I have learned different project management skills and found my shortcomings during this project. I believe that this project can be a great experience for me. In the future, I will try my best to improve my weaknesses and apply the learned skills in future projects.

Ho Sung Hin 58010837

For this project, I was in charge of portions of code writing and testing. Similar to my teammates, I found time management quite difficult, especially given the circumstances this year, we were only able to meet as a group face-to-face on Fridays. As such, most of our communication remained online, which sometimes made it harder to convey project ideas and split tasks clearly. In addition, it was also very easy to slip behind on schedule, given limited work time and the relatively short timeframe of the project. Especially during weekdays, it was tiring to make progress after daytime work. I personally felt that I had slipped behind internal deadlines at times while preparing for the practical tests and midterms for other courses. As such, although we were able to successfully deliver a final product on time, I would aim to strengthen communication and time management for future collaborative projects, making sure we try to stay on schedule as much as possible.

For the programming part, I was able to apply the knowledge from previous courses to model the interactions from different classes, as well as getting to know how to use the MVC model in a cleaner, less complicated format, which I also use in my internship placement program. In addition, I had first-hand experience with bug interactions, as we originally did not think to create unit test cases before integration, which led to us using up some additional time to debug exceptions, particularly with index issues while issuing and searching the list of player-owned cards. Although I was not in charge of creating the graphical user interface of the game, I also feel like I was able to understand more about applying the built-in packages, such as JPanel, which we did not go through in CS2312. In summary, more than programming, I was able to learn about the importance of proper team communication and scheduling, which I can apply to future team projects.